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Quantifying Recovery in Tilted Reservoirs: Insights from 2,100+ Simulation Runs on Geological, Rock, and Fluid Properties Sensitivities

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Abstract

Objectives/Scope: Understanding the relative impact of geological, rock and fluid parameters on primary oil recovery is essential for optimizing early-stage development strategies in structurally complex reservoirs. This study quantifies the first-order influence of key parameters—structural dip, horizontal permeability, vertical-to-horizontal permeability ratio (Kv/Kh), oil viscosity, driving mechanisms, relative permeabilities and initial production rate—on oil recovery factor and water cut progression. The objective is to establish a ranked sensitivity framework that supports early reservoir screening, development prioritization, and production planning.

Methods, Procedures, Process: A comprehensive sensitivity analysis was performed using 2,100+ controlled sector-model simulation runs. Each scenario isolated a single parameter, varying structural dip angles (15°, 30°, 45°), horizontal permeability (50, 150, 300 mD), Kv/Kh ratios (0.1, 0.5, 1), oil viscosities (1, 5, 10 cP), drive mechanisms (moderate/strong aquifer or water injection), and initial production rates (500-1500 stb/d). Multiple relative permeability curves were also tested. Simulations were run to a consistent production cut-off, tracking cumulative recovery factor and water cut performance over time. Tornado charts, correlation matrices, and cross-plots were used to evaluate dominant influencers and visualize performance envelopes across variable combinations.

Results, Observations, Conclusions: Dip angle, oil viscosity, and horizontal permeability emerged as the most significant factors influencing recovery and early water breakthrough. Steeper dips enhanced vertical sweep in low-viscosity, high-permeability scenarios. Low Kv/Kh ratios restricted vertical fluid movement, increasing water encroachment risk. Aquifer strength influenced both pressure support and sweep patterns. Recovery factors ranged from 6% to50%, demonstrating the wide variability driven by interacting parameters. Sensitivity rankings revealed favorable combinations and early indicators of suboptimal development conditions.

Novel/Additive Information: This study provides one of the largest parameter-isolated sensitivity datasets for tilted fault block reservoirs. Unlike typical multivariate or end-of-life-focused approaches, this workflow isolates and quantifies first-order effects in a way that is operationally relevant. By identifying which parameters most strongly control recovery performance, the results empower asset teams to prioritize appraisal efforts, fast-track promising blocks, and avoid inefficient development paths.

The methodology offers a practical, repeatable decision-support tool for screening under-characterized reservoirs and improving early resource allocation and gives robust initial estimate on recovery factor and fields performance.

Introduction

In many hydrocarbon basins around the world, especially those shaped by tectonic activity, reservoirs are seldom structurally simple. Complex structural settings—such as faulted or compartmentalized blocks and tilted stratigraphic layers—introduce significant variability in dip angles, compartment sizes, and flow geometries, all of which complicate the understanding of reservoir performance. In such environments, predicting flow behavior and assessing recovery trends become increasingly challenging due to the intricate nature of reservoir architecture and structural tilt.

These challenges are not only structural; variations in fluid and rock properties—such as oil viscosity, permeability anisotropy, and vertical-to-horizontal permeability ratios—can also significantly affect recovery performance. Even within the same structural setting, small changes in these properties may lead to notable differences in sweep efficiency, water breakthrough timing, and overall production behavior. Therefore, relying on simplified or deterministic interpretations may overlook the complex interactions between reservoir properties and fluid flow, resulting in misleading performance assessments.

While it remains common practice to benchmark recovery factors using analog fields or reservoir units with similar depositional environments or drive mechanisms, such comparisons risk inaccuracy when other critical parameters—such as dip angle, fluid viscosity, permeability distribution, and vertical connectivity—are not properly aligned. This highlights the need for a more structured, parametric benchmarking approach that accounts for the combined influence of geological structure and dynamic flow behavior.

This study addresses that gap by evaluating the results of over 2,100 simulation scenarios, covering a broad range of dip angles, rock and fluid properties, and recovery mechanisms. Through systematic variation and analysis of these parameters, the study provides a comprehensive understanding of how structural tilt and reservoir characteristics interact to influence recovery efficiency. An extended phase of this work, focused on the impact of extended shut-in periods (10, 20, and 30 years) and the incremental recovery following reactivation—including water cut behavior at re-entry—is done and will be presented in another study.

It is also important to note that all simulations in this study were conducted under identical producer well conditions, with a single production well used in each case. The resulting recovery factor values are not intended to represent field-scale or ultimate recovery, but rather to benchmark and compare how different geological and fluid settings influence the relative performance of a single well. This controlled setup enables the isolation of key trends and sensitivities, providing valuable insights for field development screening and early-stage decision-making in structurally complex reservoirs.

Methodology

This study uses a structured simulation approach to evaluate how structural dip, reservoir properties, fluid characteristics, and drive mechanisms influence oil recovery. Over 2,100 simulation runs were conducted using a consistent single-well framework, where only key parameters were varied while maintaining standardized conditions across each scenario.

Simulation Setup

A. Structural Models

Three tilted structural models were created with dip angles of 15°, 30°, and 45°, capturing a realistic range of tilts observed in structurally complex basins.

B. Grid and Model Consistency

All models were built with the same grid dimensions, cell sizes, and total number of cells to ensure uniform numerical behavior across cases.

C. STOIP Normalization

To maintain comparability, the oil-water contact (OWC) was adjusted in each structure so that all models had the same stock tank oil initially in place (STOIP).

Parameters Studied

Dip Angle. Three structural models were constructed with dip angles of 15°, 30°, and 45° to investigate the influence of gravity segregation, fluid contact orientation, and flow geometry on recovery performance.

The dip angle directly affects the direction and efficiency of fluid movement—particularly water and oil segregation—making it a critical structural parameter in tilted reservoirs. All models were built using the same grid size, dimensions, and well placement, and the OWC was adjusted in each case to maintain a consistent STOIP, enabling fair comparisons.

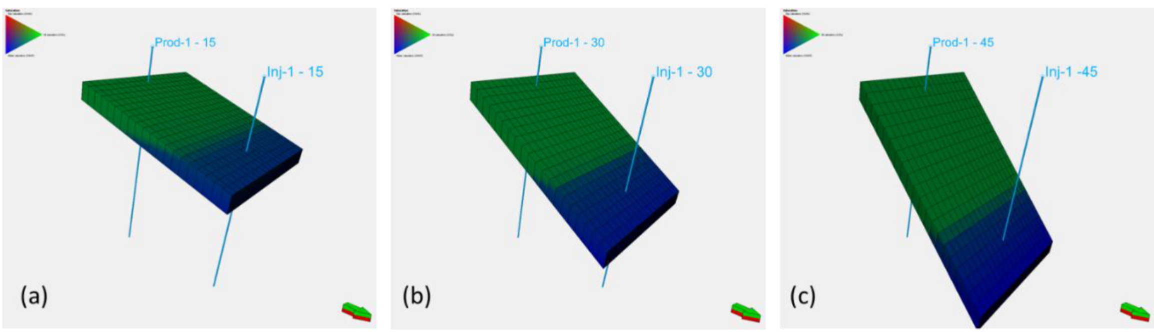

Figure 1—Different dip angles in 3 structure models with changed OWCs to have same STOIP (a) 15° dip angle; (b) 30° dip angle; (c) 45° dip angle

Horizontal Permeability (Kh). Horizontal permeability values of 50, 150, and 300 mD were selected to represent a spectrum of different quality reservoir rock. These variations were used to assess the impact of reservoir deliverability on fluid flow, areal sweep efficiency, and impact on production wells as well as injection efficiency.

Each permeability value was applied across all combinations of dip angles, vertical permeability ratios, and fluid types to isolate its role in influencing recovery dynamics. (Fig. 2)

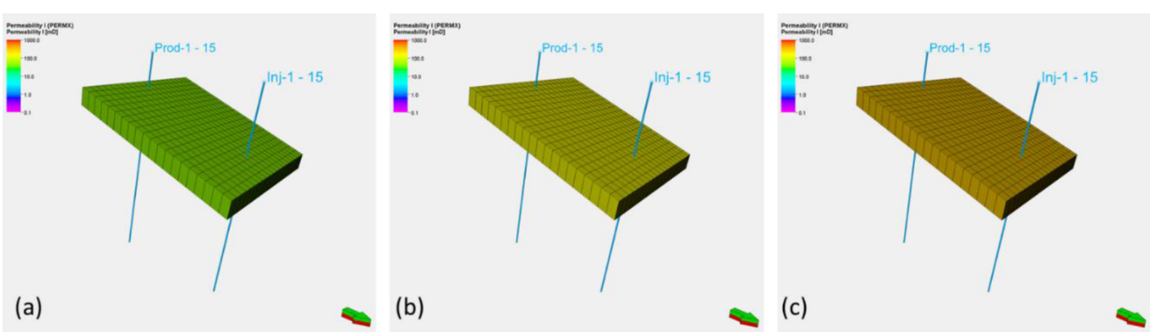

Figure 2—Different horizontal permeabilities for dip angle 15° (a) Kh = 50 mD; (b) Kh = 150 mD; (c) Kh = 300 mD

Vertical Permeability (Kv). Vertical-to-horizontal permeability ratios (Kv/Kh) of 0.1, 0.5, and 1.0 were used to evaluate vertical permeability variations effects on production rates and fluid migration under different structural conditions, as well as the water movement and water cut performance. For each horizontal permeability, the corresponding vertical permeability was varied accordingly. For example, at 50 mD horizontal permeability, the vertical permeability values used were 5 mD, 25 mD, and 50 mD, representing limited to full vertical communication. This approach allowed for consistent and structured comparison of anisotropy effects across all dip angles and flow conditions. (Fig. 3)

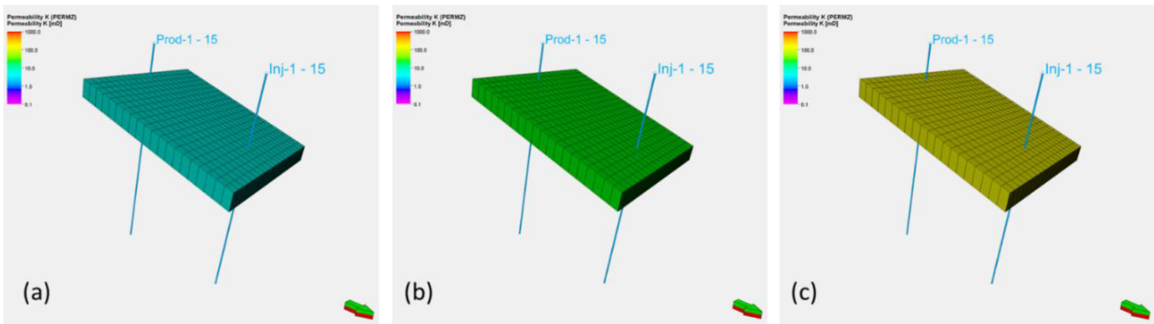

Figure 3—Different vertical permeabilities for Kv 50 mD and dip angle 15° (a) Kv= 0.1*Kh; (b) Kv= 0.5*Kh; (c) Kv= Kh

Fluid Properties – Oil Viscosity. Three distinct PVT sets were used to represent oils of varying viscosity, ranging from oil viscosity of 1 cP to relatively more viscous oil 10 cP and in between value of 5 cP. These sets were designed to capture the behavior of different fluid types encountered in clastic reservoirs. And explore how varying fluid mobility influences recovery and water break through under tilted geometry structures.

By using three distinct but realistic PVT models, the study isolates the effect of fluid type and mobility ratio across a wide range of rock properties and drive mechanisms, enabling a more robust understanding of recovery performance in structurally complex reservoirs. (Fig. 4)

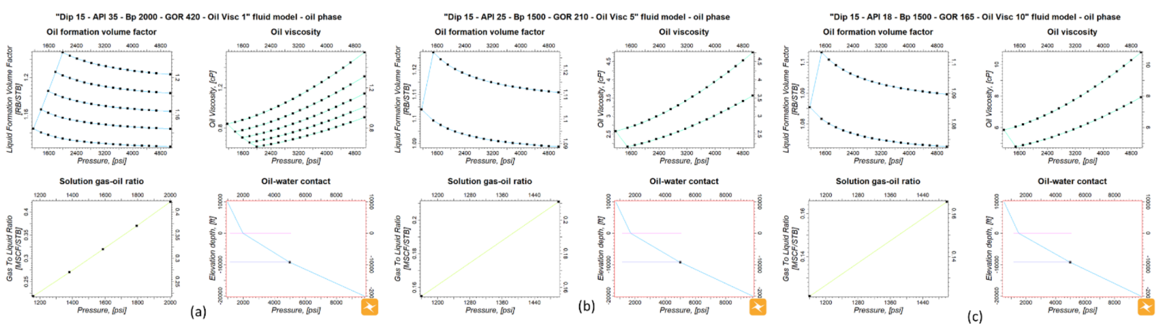

Figure 4—Different PVT models (a) Oil Visc 1 cP; (b) Oil Visc 5 cP; (c) Oil Visc 10 cP

Driving Mechanism. Three drive mechanisms were modeled to evaluate recovery behavior under different energy support systems: a moderate aquifer, a strong aquifer, and water injection. All scenarios were configured using the same structural and grid models to ensure consistency and allow direct comparison.

The aquifer models were constructed using the Carter–Tracy bottom-water drive model, with variations in permeability, thickness, and external radius to represent moderate and strong pressure support. These parameters were adjusted to ensure a controlled increase in aquifer strength while maintaining the same overall configuration across cases.

In the water injection scenarios, injection commenced from day one of production, with a voidage replacement ratio (VRR) of 1.0, to maintain reservoir pressure. Injector locations, rates, and operational settings were standardized across all water injection runs.

As shown in the below (Fig. 5) three simulation runs with the same dip angle 15°, Kh 50mD, Kv 5mD, Oil Visc 1 cP, Initial Rate of 1500 bopd with the three different driving mechanisms, it show difference in pressure behavior, water cut % and Recovery factor %.

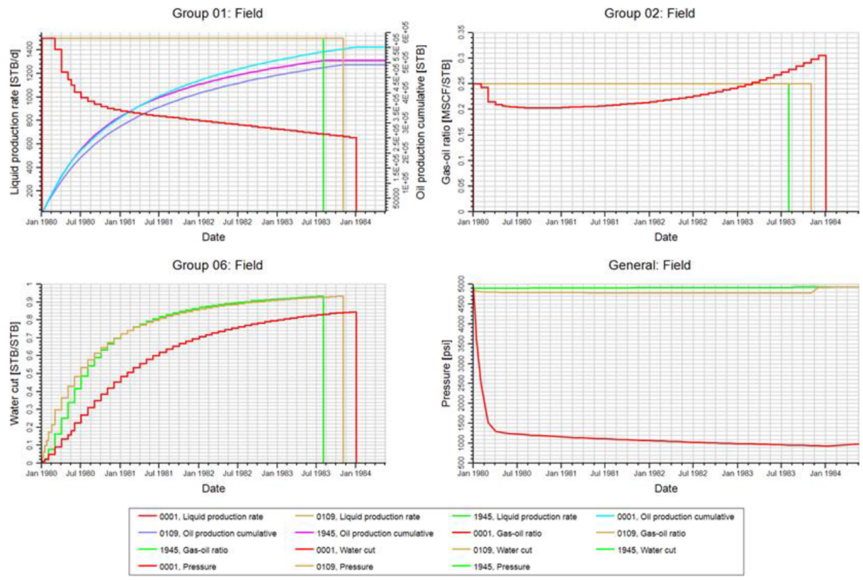

Figure 5—Different Driving mechanism Moderate Aquifer - Red; (b) Strong Aquifer - Brown; (c) Water Injection – Green

Wettability (Relative Permeability Curves). To evaluate the impact of wettability on displacement efficiency and recovery behavior, two distinct relative permeability curves were used across the simulation cases. These curves represent different wettability behaviors by varying key characteristics for rock physics such as Corey Oil and Corey Water.

Variations in Corey exponents influence the shape of the relative permeability curves, thereby affecting oil mobility, water advance, and the overall sweep pattern. The use of two representative curves enabled a comparative analysis of how subtle changes in rock–fluid flow characteristics can influence recovery under different structural and operational conditions.

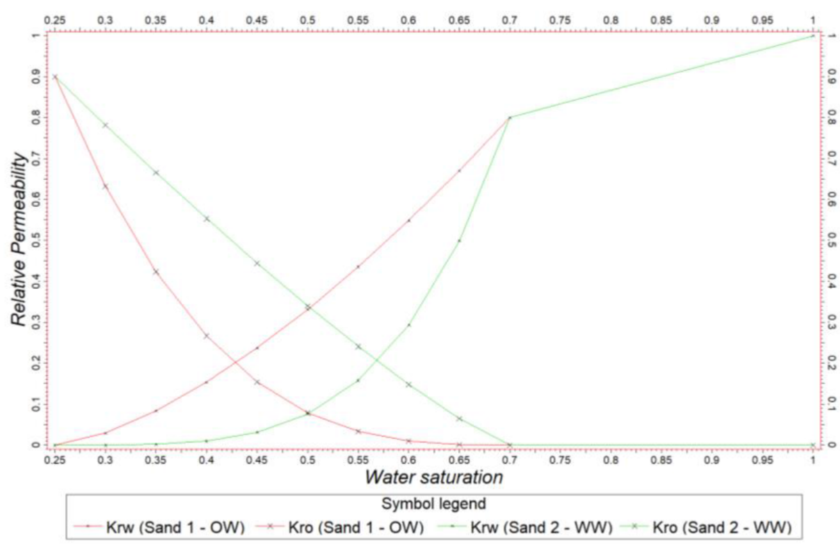

Figure 6—Different Relative Permeability Curves Kr Curve 1 – Red; (b) Kr Curve 2 - Green

Initial Production Rate. Two distinct initial production rates (500-1500 bopd) were applied across the simulation cases to evaluate the impact of early production strategy on recovery performance. The two rate levels were selected to represent aggressive and conservative production scenarios, enabling a focused sensitivity analysis on how the choice of starting rate affects recovery factor, pressure behavior, and water breakthrough timing.

This approach highlights how early production planning—particularly in structurally complex reservoirs—can influence reservoir dynamics and long-term recovery outcomes.

In below (Fig. 7) two identical runs, dip angle 45°, Kh 300 mD, Kv 150 mD, Oil Visc 5 cP, Kr curve 1 and moderate aquifer with two different production strategies, conservative 500 bopd and aggressive 1500 bopd, the conservative production strategy show slower WC% and GOR trends, and lower pressure depletion, consequently higher recovery factor at late shut-in time, This illustrates the critical influence of early production strategy on long-term reservoir performance, even in otherwise identical reservoir conditions.

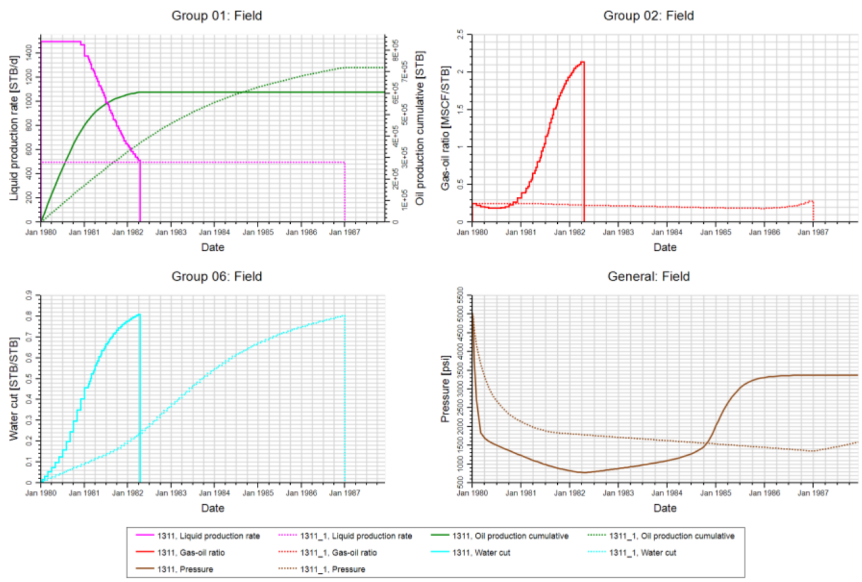

Figure 7—Different Initial Rates Oil Rate 1500 – solid lines; (b) Oil Rate 500 – dashed lines

Simulation Design Notes

- STOIP is fixed constant in all cases by changing OWC with different dip angles within the three structural models.
- Each simulation case included a single vertical producer, placed at the crest of the reservoir structure. The recovery factor calculated from each run reflects the relative performance of the system under the given conditions. These are not intended to represent full-field or ultimate recovery estimates.
- All producer wells are produced under same operating conditions, as well as injection criteria in case of water injection operating conditions are fixed.
- A parametric approach was followed where only one variable was changed at a time, allowing for clear attribution of performance changes to individual parameters.
- Some parameter combinations were not simulated for the 30° dip angle, as early screening showed they would yield limited additional insight. This allowed computational resources to focus on the most impactful and diverse scenarios

Results

To evaluate the effect of each parameter on reservoir performance, a systematic sensitivity analysis was performed using a one variable at a time approach. In each comparison, one parameter was varied while all other inputs were held constant, allowing for clear isolation of its individual impact. Oil production performance, WC% performance, GOR performance, Cumulative oil production and recovery factor %, and water saturation distribution in 3D are the main key performance parameters that are observed in every run.

Dip Angle. Dip angle directly impacts the fluid movement and gravity segregation as shown in the below 3D screenshots from the model comparing saturation distribution and production performance difference between two runs having same parameters of Kh = 300 mD, Kv = 30 mD, Oil Viscosity 1 cP, moderate strength and initial rate of 1500 bopd, the only parameter changed is the dip angle between 15-45° (Fig. 8)

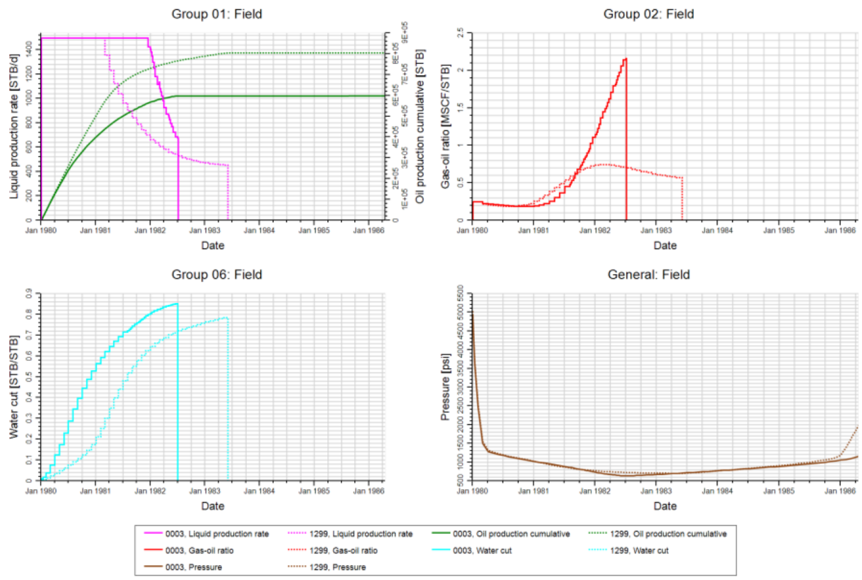

Figure 8—Different dip angles production performance Dip angle 15° – solid lines; (b) Dip angle 45 – dashed lines

It's observed that the 15° dip angle results in faster water movement towards the producer well hence earlier water breakthrough while in the 45° dip angle slower water movement hence more oil produced, for the GOR trends in the 45° dip angle it's observed that the GOR trend is much less than the 15° dip angle, as more gas saturation is observed in the case of 15° dip angle around the producer, hence the well produces with higher GOR

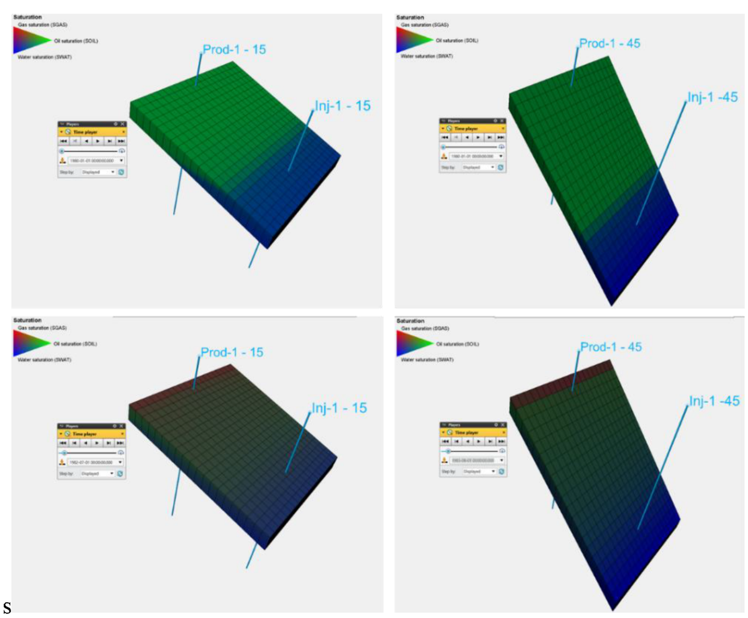

Figure 9—Saturation distribution at different dip angles Dip angle 15°; (b) Dip angle 45°

In the chart below (Fig. 10) is comparison between 15°, 30° and 45° dip angles and it shows consistently that the recovery factor is directly proportional to dip angle, as the dip angle increases, the water breakthrough is delayed allowing more oil to be recovered. So when taking analogy of recovery factor from nearby field, if the dip angle is slightly higher, therefore recovery factor should be enhanced by different factors according to the case and the RF% value itself.

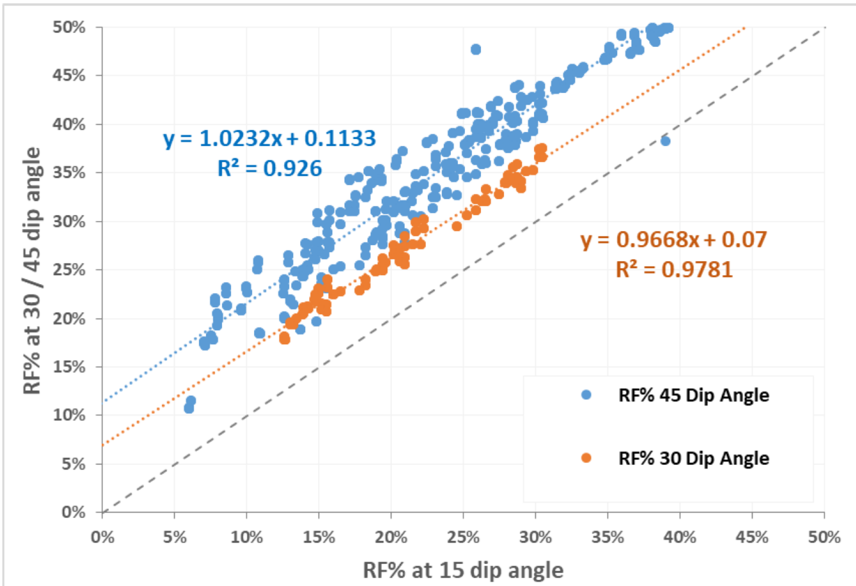

Figure 10—Recovery factor % vs different dip angles

Horizontal Permeability. Horizontal permeability plays a key role in determining reservoir deliverability and sweep efficiency. To analyze its effect, 3 runs where the only difference is the Kh from 50 mD, 150 mD and 300 mD, where all the other parameters are fixed Dip angle = 30°, Kv = 0.5 Kh, Oil Viscosity 5 cP, moderate strength and initial rate of 1500 bopd. (Fig. 11)

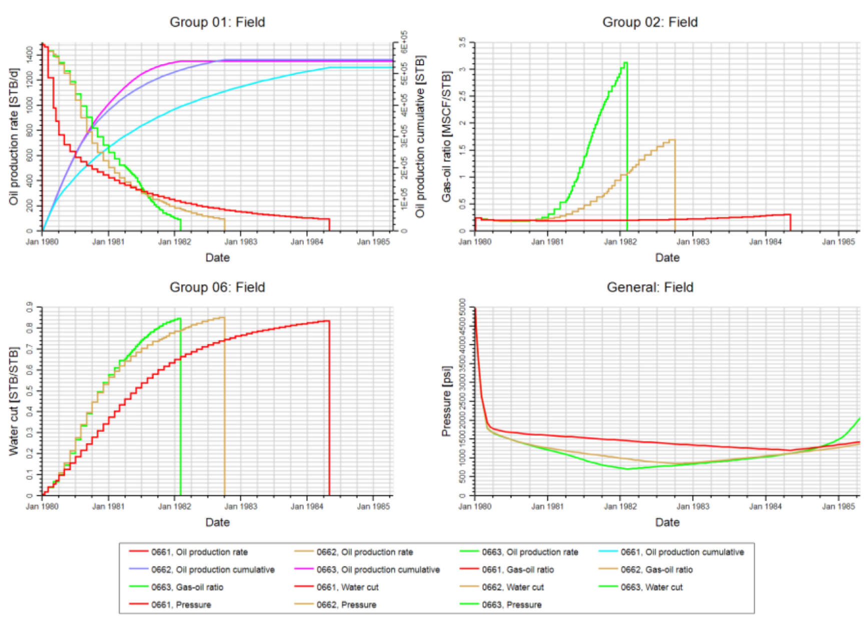

Figure 11—Different Kh production performance Kh = 50 mD – red lines; (b) Kh = 150 mD - brown lines (c) Kh = 300 mD – green lines

Despite the notable impact of horizontal permeability on flow rates and fluid movement, the final recovery factors across the three permeability scenarios (50, 150, and 300 mD) were found to be relatively close at the end of the simulation runs. Higher permeability cases enabled the wells to produce at higher rates with earlier and sharper water breakthrough, resulting in accelerated oil recovery. However, these case also exhibited higher GOR trends due to faster gas coning near the wellbore in the moderate aquifer (Fig. 12), in the same conditions with injection support, there'll be no huge difference in the overall performance – for this range of permeability used in this study.

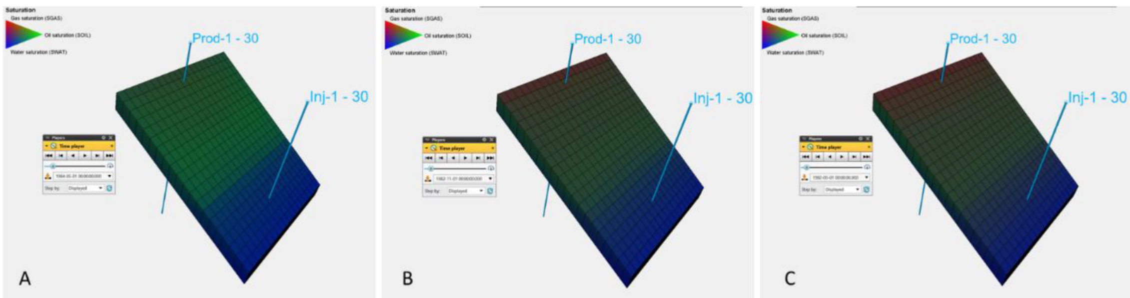

Figure 12—Saturation distribution at different vertical permeabilities Kh = 50 mD; (b) Kh = 150 mD (c) Kh = 300 mD

In the chart below (Fig. 13), recovery factors are compared across three horizontal permeability values: 50, 150, and 300 mD. The results indicate that while horizontal permeability does have an impact on the ultimate recovery factor, the impact is relatively moderate.

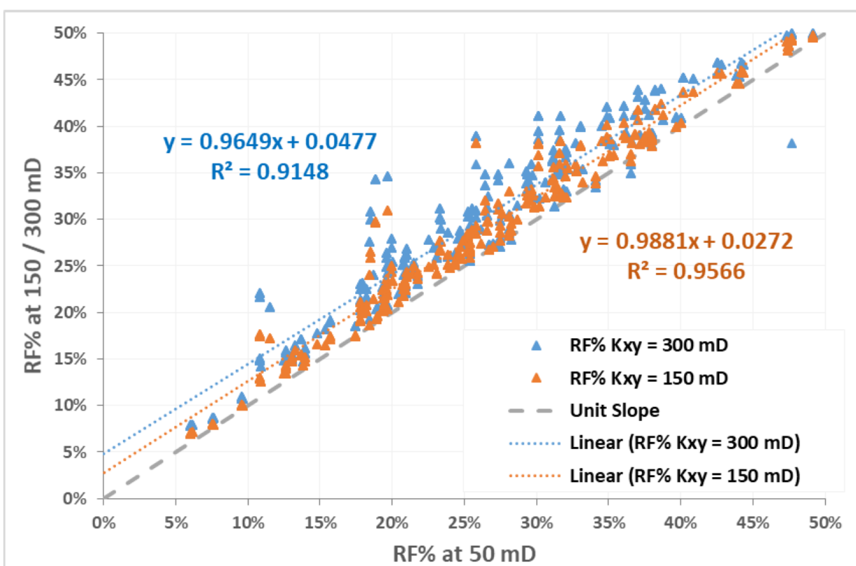

Figure 13—Recovery factor % vs horizontal permeability values

However, when the time to recovery is considered, the influence becomes much more significant — with higher permeability accelerating oil production and improving recovery efficiency within development time.

Vertical Permeability. Vertical permeability plays a key role in governing vertical fluid communication, gravity segregation, and the advancement of water cut and gas-oil ratio (GOR) trends, especially in tilted reservoirs and weaker aquifer support. In this study, vertical-to-horizontal permeability ratios (Kv/Kh) of 0.1, 0.5, and 1.0 were evaluated across multiple scenarios to assess their influence on recovery.

While changes in vertical permeability clearly influenced the timing of water and gas movement, the overall impact on final recovery factor was relatively minor across the simulation spectrum. To analyze its effect, 3 runs where the only difference is the (Kv/Kh) from 0.1, 0.5 and 1, where all the other parameters are fixed Dip angle = 45°, Kh = 300 mD, Oil Viscosity 1 cP, moderate strength and initial rate of 1500 bopd. (Fig. 14)

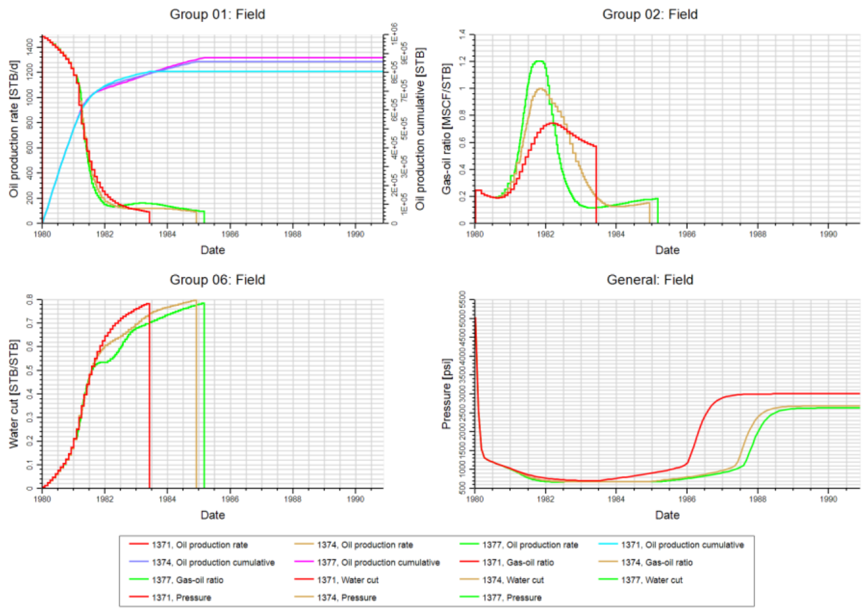

Figure 14—Different Kv/Kh production performance Kv/Kh = 0.1 – red lines; (b) Kv/Kh = 0.5 - brown lines (c) Kv/Kh = 1 – green lines

In the three cases evaluated under a moderate aquifer, the GOR trend was highest for the case with Kv/Kh = 1.0, indicating enhanced gas mobility due to stronger vertical connectivity. However, under other drive mechanisms such as strong aquifer or water injection, these differences were far less significant. The support from stronger pressure maintenance appeared to compensate for variations in vertical permeability, resulting in minimal differences in production behavior or recovery factor values across the tested Kv/Kh range.

However drawing the RF% of Kv/Kh of 0.5 and 1 vs RF% of Kv/Kh 0.1 show that Kv/Kh have minor impact on total recovery factor % (Fig. 15)

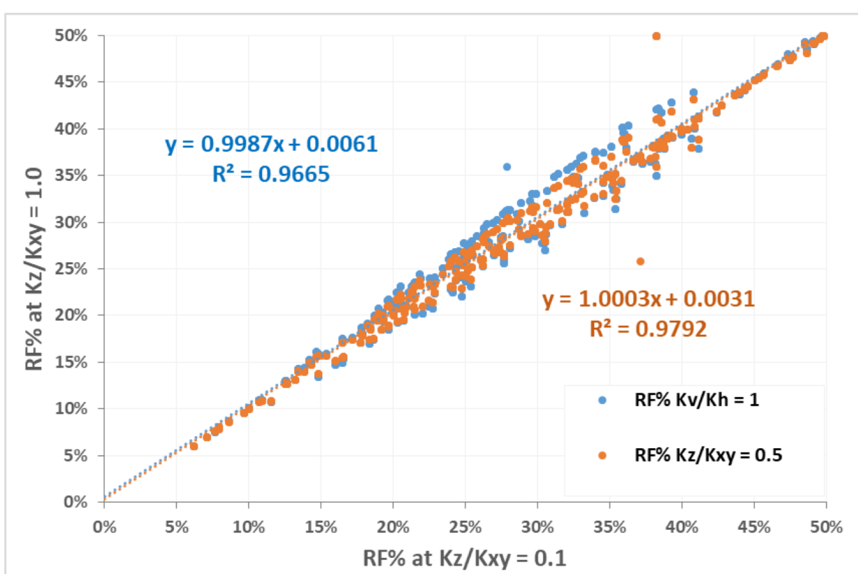

Figure 15—Recovery factor % vs vertical permeability values

Fluid Properties – Oil Viscosity. Oil viscosity directly influences fluids mobility, sweep efficiency, and the timing of water and gas breakthrough. To isolate this effect, three distinct PVT models were constructed with oil viscosities of 1 cP, 5 cP, and 10 cP, enabling controlled variation of mobility ratios.

By using these three PVT sets, the simulations captured how differences in mobility ratio affect fluid front advancement, pressure behavior, and recovery performance under otherwise identical conditions. To analyze its effect, 3 runs where the only difference is the Oil Visc from 1 cP, 5 cP and 10 cP, where all the other parameters are fixed Dip angle = 15°, Kh = 50 mD, Kv = 5 mD, strong aquifer strength and initial rate of 500 bopd. (Fig. 16)

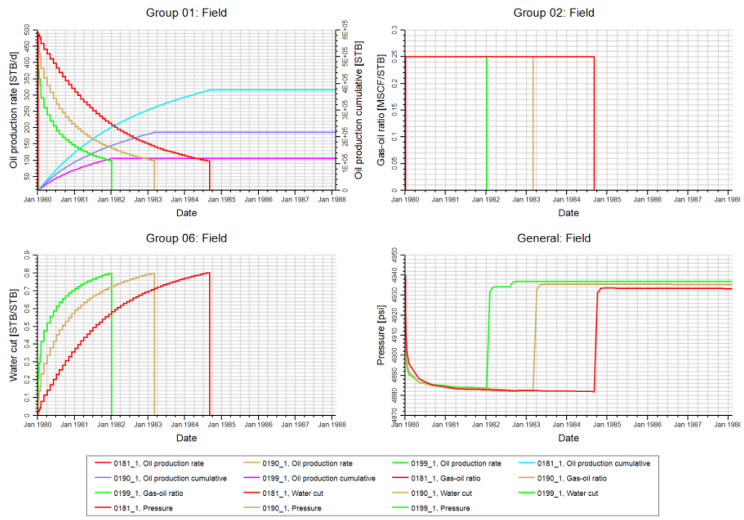

Figure 16—Different oil viscosity production performance Oil Visc = 1 cP – red lines; (b) Oil Visc = 5 cP - brown lines (c) Oil Visc = 10 cP – green lines

The results clearly demonstrate the role of mobility ratio in shaping recovery performance across the three fluid cases. As oil viscosity increases and mobility decreases, displacement becomes less efficient, leading to earlier water breakthrough and lower recovery under otherwise identical conditions. Heavier oil systems consistently yield reduced recovery factors, reinforcing the importance of fluid characterization in forecasting reservoir performance.

In the snapshots presented in (Fig. 17), the water saturation distribution in reservoir layer 10 is shown for each viscosity scenario, captured one month after well shut-in. This saturations highlight the increasing severity of water coning and upward encroachment with rising oil viscosity. Higher-viscosity fluids exhibit lower mobility ratios, resulting in less stable displacement fronts, earlier water breakthrough, reduced oil drainage efficiency, and ultimately lower volumetric recovery. These outcomes illustrate how adverse mobility conditions can accelerate water dominance near the wellbore, particularly in structurally tilted systems where gravity-driven segregation and fluid viscosity interact.

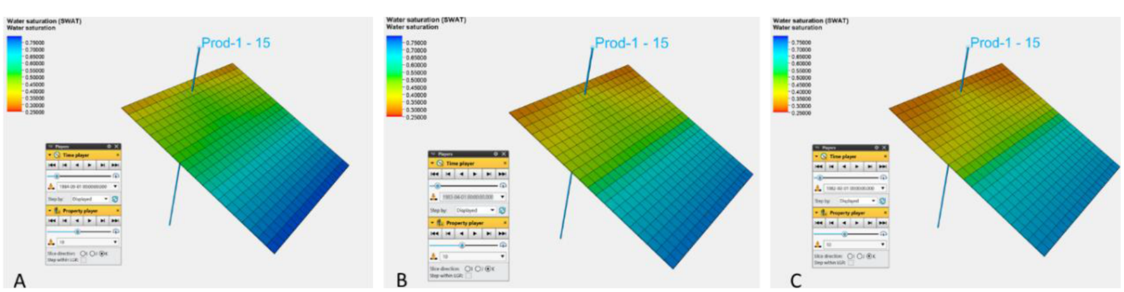

Figure 17—Saturation distribution at different oil viscosities Oil Visc = 1 cP; (b) Oil Visc = 5 cP (c) Oil Visc = 10 cP

Plotting the recovery factor at 1 cP viscosity against the values at 5 cP and 10 cP provides a clear indication of the extent to which fluid properties alone can influence recovery performance under identical reservoir and operational conditions. The cross plot in (Fig. 18) highlights the progressive reduction in recovery as viscosity increases, emphasizing the critical role of fluid mobility in determining sweep efficiency and ultimate recovery.

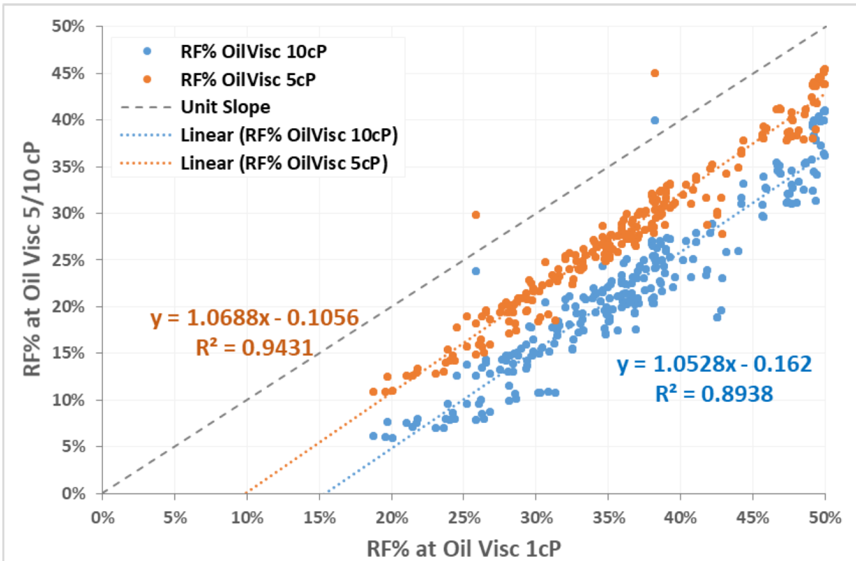

Figure 18—Recovery factor % vs different oil viscosity values

Driving mechanisms. Different driving mechanism impact on recovery factor is clear and well-known in the industry the impact of having moderate, strong aquifer on recovery, or even water injection. However to put this factor into consideration, it was tested as well.

To analyze its effect, shown below three runs where the only difference is the driving mechanism moderate aquifer, strong aquifer or water injection, where all the other parameters are fixed Dip angle = 30°, Kh = 300 mD, Kv = 150 mD, Oil Viscosity = 1 cP and initial rate of 1500 bopd. (Fig. 19)

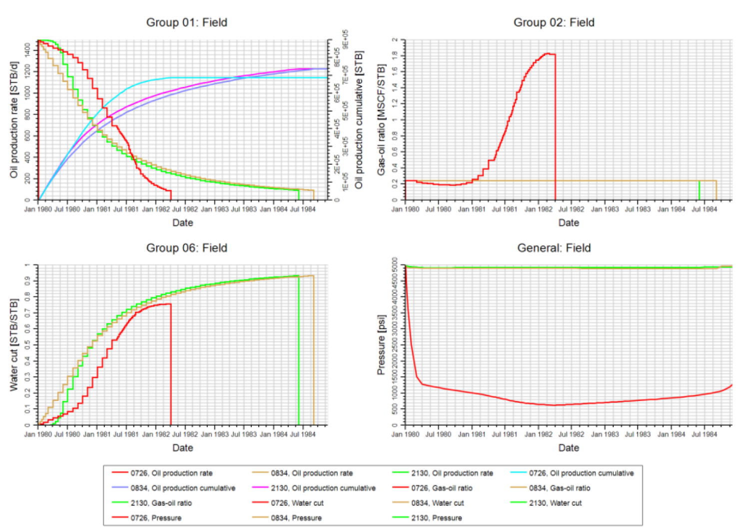

Figure 19—Different driving mechanism production performance Moderate aquifer – red lines; (b) Strong Aquifer – brown lines (c) Water Injection – green lines

While differences in production behavior across drive mechanisms are evident—particularly in pressure trends and breakthrough timing—the variation in ultimate recovery factor remains relatively limited in this study. This is primarily because the moderate aquifer cases do not reach reservoir abandonment limits, and reservoir pressure remains above critical drawdown thresholds, allowing producers to remain active and achieve comparable recovery to other support strategies.

Furthermore, the broad range of oil viscosities (1, 5 and 10 cP) introduces contrasting mobility ratios, which influence displacement front stability, sweep efficiency, and fluid segregation. In low-viscosity cases, gravity-driven segregation and moderate pressure support enable efficient oil drainage. Conversely, in higher-viscosity cases, mobility-dominated flow resistance reduces the effectiveness of pressure support, and more aggressive strategies—such as increased offtake rates or polymer injection—may be required to enhance recovery.

As seen in the below cross-plots, the enhancement in RF% from moderate to strong aquifer. (Fig. 20a) and enhancement in RF% from moderate to water injection. (Fig. 20b)

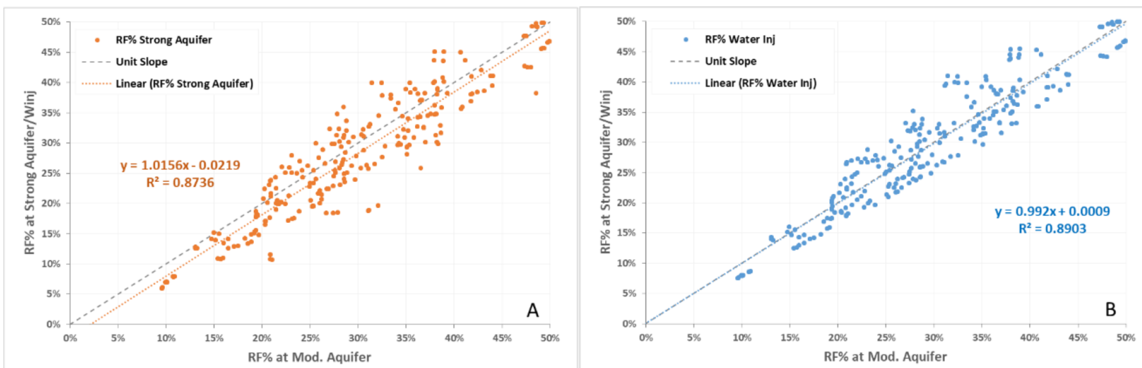

Figure 20—Recovery factor % vs driving mechanisms Moderate vs Strong Aquifer; (b) Moderate Aquifer vs Water Injection

If considered the time factor in different recovery mechanisms, it's found that stronger aquifer support and water injection mechanism can enable more operating time in most cases while fewer cases show less operating time as shown in (Fig. 21), showing months online and recovery factor comparison between moderate aquifer to strong aquifer (Fig. 21A) and moderate aquifer to water injection (Fig. 21B)

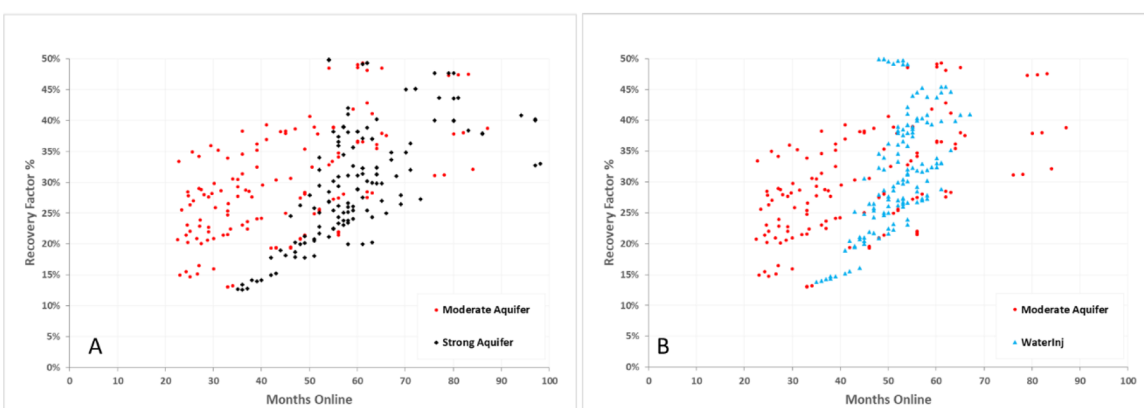

Figure 21—Months online vs recovery factor % for different driving mechanisms (a) Moderate vs Strong Aquifer; (b) Moderate Aquifer vs Water Injection

Wettability (Relative Permeability Curves). Rock-fluid interaction, particularly wettability, plays a critical role in governing the relative movement of oil and water within the reservoir. This is primarily captured through relative permeability curves, which define the phase mobility as a function of saturation. Oil-wet and water-wet systems exhibit fundamentally different flow behavior and displacement efficiencies due to differences in capillary forces, fluid distribution, and residual saturations.

To evaluate this effect, two distinct relative permeability (Kr) curves were introduced, as shown in (Fig. 6). Two different Corey-type relative permeability curves were used by varying the Corey exponents for oil and water. The oil-wet case (Kr Curve 1) has higher Corey exponents, resulting in slower mobility increase and delayed water breakthrough. The water-wet case (Kr Curve 2) uses lower exponents, allowing earlier water movement and more rapid displacement. To analyze its effect, 2 runs where the only difference is the relative permeability curves, where all the other parameters are fixed Dip angle = 15°, Kh = 150 mD, Kv = 15 mD, Oil Viscosity = 1 cP and initial rate of 1500 bopd. (Fig. 22)

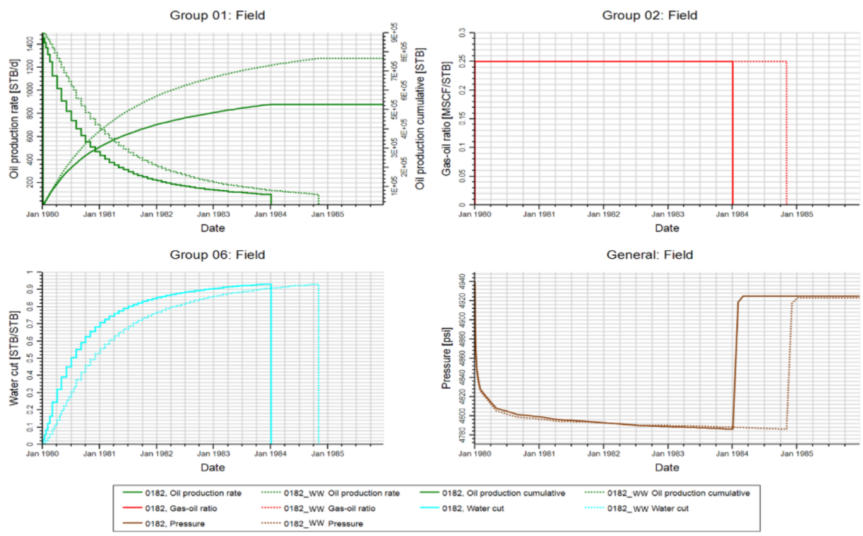

Figure 22—Different relative permeability curves production performance Kr curve 1 – Oil wet – Solid lines; (b) Kr curve 2 – Water wet – Dashed lines

As shown in the cross-plot below (Fig. 23), the difference in recovery factor between Kr Curve 1 – Oil wet and Kr Curve 2 – Water wet highlights the significant impact that rock-physics functions—particularly relative permeability behavior—can have on recovery. The enhanced recovery observed with Kr Curve 1 reflects improved phase mobility and displacement efficiency under the same reservoir conditions.

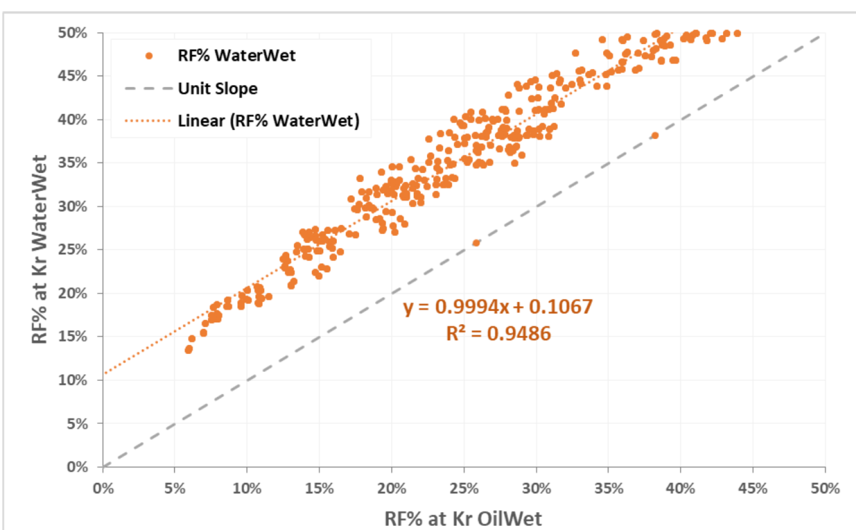

Figure 23—Recovery factor % vs different relative permeability curves

Initial Production Rate. Initial production rates can influence pressure drawdown, fluid mobility, and the timing of water and gas breakthrough, especially in reservoirs with limited pressure support or gravity segregation. To evaluate this effect, two distinct production scenarios were tested, conservative case 500 bopd and aggressive case 1500 bopd initial rates.

The sensitivity analysis on initial production rates revealed that reservoir performance is strongly influenced by the level of pressure support. In cases with moderate aquifer strength (Fig. 24A), lower offtake rates help preserve reservoir pressure, delay water breakthrough, and maintain favorable mobility conditions, resulting in higher recovery factor. Conversely, in scenarios with strong aquifer support (Fig. 24B) or water injection, aggressive production enhances oil recovery. The pressure support compensates for the higher drawdown, improving recovery efficiency.

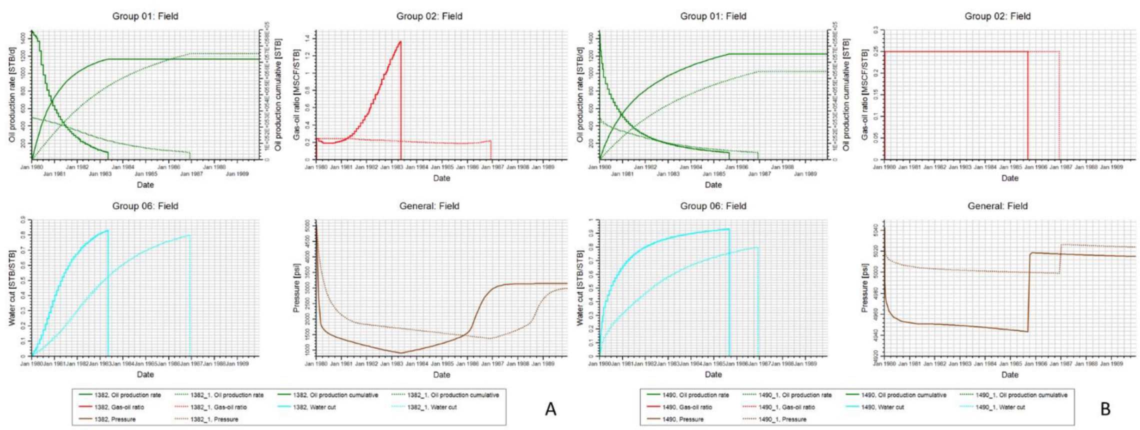

Figure 24—Different production strategies production performance [A- Moderate aquifer B- Strong Aquifer] Qi=1500 bopd – Solid lines; (b) Qi=500 bopd – Dashed lines

The cross-plot in (Fig. 25) compares recovery factors between aggressive (Qi = 1500 BOPD) and conservative (Qi = 500 BOPD) production strategies, with points color (hue) representing the drive mechanism. The plot shows that most moderate aquifer cases (yellow points) fall below the unit-slope line, indicating better performance under conservative production strategies. In contrast, cases with strong aquifer support or water injection tend to lie above the unit-slope line, suggesting improved recovery under aggressive production strategies where there's enough pressure support compensates for drawdown.

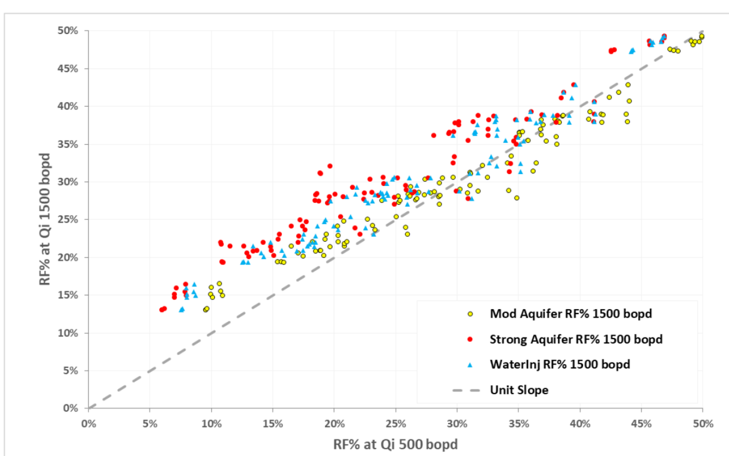

Figure 25—Recovery factor % vs different initial rates (Hues – driving mechanisms)

Conclusions and discussions

This paper presented a parametric study to evaluate factors having impact on recovery factor performance. The analysis systematically isolated the impact of seven key parameters — dip angle, horizontal and vertical permeability, oil viscosity, drive mechanism, relative permeability curve, and initial production rate — under consistent reservoir architecture and operating conditions.

Fig. 26shows that oil viscosity has the most impact on recovery factor and vertical permeability has the least impact.

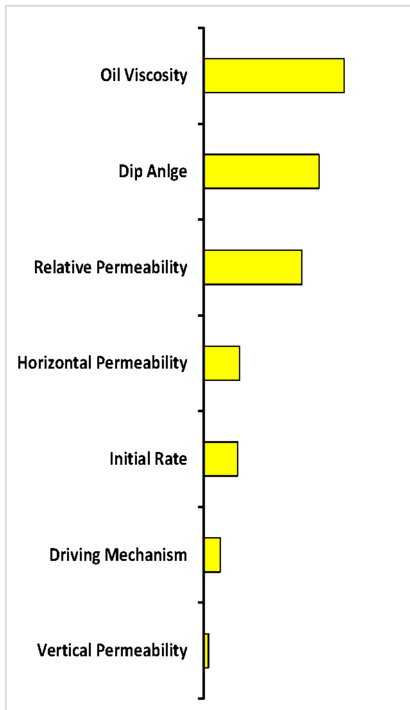

Figure 26—Tornado chart for factors impacting recovery factor

Key Observations

1. Oil Viscosity: showed a clear inverse relationship with recovery: lower-viscosity fluids yielded better displacement and higher recovery factor
2. Dip Angle: strong impact on recovery factor, with higher dip angle enhancing gravity segregation, delaying water breakthrough, and improving sweep efficiency hence recovery factor.
3. Wettability: moving from water wet to oil wet relative permeability curves has huge impact on recovery and production performance.
4. Horizontal permeability: had a moderate impact on the final recovery factor when the time dimension was excluded. However, higher permeability cases enabled earlier recovery of the same volumes. Which may have significant implications in economic.
5. Production strategies: (Initial rate) played an important role in recovery outcomes, particularly under moderate aquifer support. Conservative production rates delayed water and gas breakthrough, improving displacement efficiency and resulting in higher ultimate recovery. In contrast, aggressive production led to faster pressure depletion and earlier encroachment. However, under strong aquifer support, higher production rates were more sustainable and had minimal negative impact on recovery, suggesting that the optimal production strategy is highly dependent on the reservoir's pressure support system.

6. Driving mechanisms: Drive mechanism had a moderate impact on recovery, with water injection and strong aquifer yielding slightly higher RF% than moderate aquifer support in some cases. This limited difference is attributed to the long simulation time, balanced VRR injection, and gravity-dominated flow in the single-well, tilted reservoir setup, as well as the well constraints.
7. Vertical permeability: had limited impact on recovery despite its impact on the fluid movement, WC % and GOR trends.

Another dimension of this study involved extended shut-in and reactivation cycles. After each well reached its operating limits (uniform across all runs), it was shut in and later reactivated following shut-in durations of 10, 20, and 30 years. This approach allowed assessment of gravity-driven fluid redistribution over time and its effect on initial water cut, second-phase production performance, and incremental oil recovery. By repeating this process across the full range of parameters, the study captures the combined influence of reservoir properties, structure, and time-dependent gravity segregation on post-shut-in recovery potential.

This extended analysis provides valuable insights for re-accessing bypassed oil in long-inactive fields, with some cases showing incremental recovery improvements of up to 10%, reinforcing the potential value of late-life reactivation strategies.

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